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# DISASTER PREPAREDNESS AND RESPONSE EDUCATION SYSTEM
# Sample Backend Implementation using Flask + MySQL
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from flask import Flask, render_template, request, redirect, session, url_for
import mysql.connector

app = Flask(__name__)
app.secret_key = "disaster_edu_secret_key"

# -----
# Database Connection
# -----
def get_db_connection():
    conn = mysql.connector.connect(
        host="localhost",
        user="root",
        password="password",
        database="disaster_edu_db"
    )
    return conn

# -----
# Home Route
# -----
@app.route('/')
def home():
    return render_template('index.html')

# -----
# User Registration
# -----
@app.route('/register', methods=['GET', 'POST'])
def register():
    if request.method == 'POST':
        name = request.form['name']
        email = request.form['email']
        password = request.form['password']
        role = request.form['role']

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conn = get_db_connection()
cursor = conn.cursor()
cursor.execute(
    "INSERT INTO users (name, email, password, role) VALUES (%s, %s, %s, %s)",
    (name, email, password, role)
)
conn.commit()
cursor.close()
conn.close()
return redirect(url_for('login'))

return render_template('register.html')

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# -----
# User Login
# -----
@app.route('/login', methods=['GET', 'POST'])
def login():
    if request.method == 'POST':
        email = request.form['email']
        password = request.form['password']

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conn = get_db_connection()
cursor = conn.cursor(dictionary=True)
cursor.execute(
    "SELECT * FROM users WHERE email=%s AND password=%s",
    (email, password)
)
user = cursor.fetchone()
cursor.close()
conn.close()

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if user:
    session['user_id'] = user['user_id']
    session['role'] = user['role']
    if user['role'] == 'admin':
        return redirect(url_for('admin_dashboard'))
    return redirect(url_for('dashboard'))
else:
    return "Invalid credentials, please try again."

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return render_template('login.html')

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# -----
# Student Dashboard - Learning Modules
# -----
@app.route('/dashboard')
def dashboard():
    if 'user_id' not in session:
        return redirect(url_for('login'))

    conn = get_db_connection()
    cursor = conn.cursor(dictionary=True)
    cursor.execute("SELECT * FROM modules")
    modules = cursor.fetchall()
    cursor.close()
    conn.close()

    return render_template('dashboard.html', modules=modules)

# -----
# Quiz Module
# -----
@app.route('/quiz/<int:module_id>', methods=['GET', 'POST'])
def quiz(module_id):
    conn = get_db_connection()
    cursor = conn.cursor(dictionary=True)
    cursor.execute("SELECT * FROM quizzes WHERE module_id=%s", (module_id,))
    questions = cursor.fetchall()

    if request.method == 'POST':
        score = 0
        for q in questions:
            selected = request.form.get(f"q{q['quiz_id']}")
            if selected == q['correct_answer']:
                score += 1

        total = len(questions)
        percentage = (score / total) * 100 if total > 0 else 0

    # Save progress
    cursor.execute(
        "INSERT INTO progress (user_id, module_id, status, score) VALUES (%s, %s, %s,

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%s)",
(session['user_id'], module_id, 'completed', percentage)
)
conn.commit()

# Award badge if score is high
if percentage >= 80:
    cursor.execute(
        "INSERT INTO badges (user_id, badge_name) VALUES (%s, %s)",
        (session['user_id'], 'Safety Champion')
    )
    conn.commit()

cursor.close()
conn.close()
return render_template('result.html', score=score, total=total,
percentage=percentage)

cursor.close()
conn.close()
return render_template('quiz.html', questions=questions, module_id=module_id)

# -----
# Mock Scenario Simulation
# -----
@app.route('/scenario/<disaster_type>')
def scenario(disaster_type):
    scenarios = {
        "earthquake": "The ground starts shaking. What do you do first?",
        "fire": "You smell smoke in the hallway. What is your first action?",
        "flood": "Water levels are rising rapidly near your school. What should you do?"
    }
    question = scenarios.get(disaster_type,

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